

Olga Balsalobre-Ruza

PhD candidate [defending in May 2026]

Curriculum Vitæ

Center for Astrobiology (CAB)

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[See personal webpage](#)

Spanish [Native] & English [Advanced, C1]

I hold a degree in Physical Sciences with a major in Astrophysics from the Complutense University of Madrid. During my studies, I did internships across diverse areas of astrophysics that helped me build both technical skills and scientific independence, including high-energy astrophysics, observational astronomy, and exoplanetary science.

Currently, I am a fourth-year PhD student at the Center for Astrobiology in Madrid, specializing in the **search and characterization of exoplanets and co-orbital systems** using multiple techniques, with a strong focus on **radial velocities** (RV) and **direct imaging** (DI). I have designed observational strategies for RV surveys and gained extensive experience with both ground- and space-based observatories, being the **PI of 8 approved observing programs**. I am an enthusiastic Python programmer with a keen interest in expanding my skills, for instance in machine learning and astro-statistics.

During my PhD, I have published **4 first-author** and **7 co-author papers**, developed **2 Python packages**, and presented my research at international conferences and institutions (**16 talks** and **6 posters**). I have also communicated my work to the public through several media. I thrive in both independent and collaborative research environments, valuing teamwork as essential for scientific progress.

1 Education

Bachelor in Physics UCM	BSc thesis Gamma Rays data analysis detected with Cherenkov telescopes (CTA)	Master in Astrophysics UCM	MSc thesis Development of KOBESim [see coding Section]	PhD in Astrophysics UCM & CAB	PhD thesis Search for co-orbital exoplanets via RV, DI & transits (T)
- Honors in three subjects - Speciality in Astrophysics	2015 - 2020	- Honors in two subjects - Honors in the MSc thesis	2020 - 2021	Supervised by Dr. Jorge Lillo-Box & Dr. Nuria Huélamo	2022 - [May 2026]

2 Publications in peer-reviewed journals

The observational technique used is shown after the hyphen as – **RV, DI, T**.

First author

- (5) **Balsalobre-Ruza et al.**, in prep. – **DI**
Inner dust clumps in PDS 70 revealed by SPHERE with star-hopping
- (4) **Balsalobre-Ruza et al.**, 2025 [A&A, 694, A15] – **RV & T**
KOBE-1: The first planetary system from the KOBE survey. Two planets likely in the sub-Neptune mass regime around a late K dwarf
- (3) **Balsalobre-Ruza et al.**, 2024 [A&A, 689, A53] – **RV & T**
The TRÖY project. III. Exploring co-orbitals around low-mass stars
- (2) **Balsalobre-Ruza et al.**, 2023 [A&A, 675, A172] [[ESO press release](#)] – **DI**
Tentative co-orbital submillimeter emission within the Lagrangian region L5 of the protoplanet PDS 70 b
- (1) **Balsalobre-Ruza et al.**, 2022 [A&A, 669, A18] – **RV**
KOBESim: A Bayesian observing strategy algorithm for planet detection in radial velocity blind-search surveys

Co-author

- (7) Grouffal et al., 2025 [Accepted in A&A, arXiv:2507.01807] – **RV**
The star HIP 41378 potentially misaligned with its cohort of long-period planets
Contribution: Analysis and text on CARMENES RV, Sect. 2 (author 9/32)
- (6) Figueira et al., 2025 [A&A, 700, A174] – **RV**
A comprehensive study on radial velocity signals using ESPRESSO: Pushing precision to the 10 cm/s level
Contribution: Scientific feedback on photometric centroid analysis (author 9/33)
- (5) Cesario et al., 2024 [A&A, 692, A172] – **DI**
Large Interferometer For Exoplanets (LIFE). XIV. Finding terrestrial protoplanets in the galactic neighborhood
Contribution: Manuscript revision (author 27/64)

- (4) Lillo-Box et al., 2024 [A&A, 689, L8] — **RV & T**
K2-399 b is not a planet: The Saturn that wandered through the Neptune desert is actually a hierarchical eclipsing binary
 Contribution: Manuscript revision (author 8/22)
- (3) Lillo-Box et al., 2024 [A&A, 686, A232] — **DI**
The AstraLux-TESS high spatial resolution imaging survey. Search for stellar companions of 215 planet candidates from TESS
 Contribution: Scientific feedback on photometric centroid analysis (author 4/9)
- (2) Christiaens et al., 2023 [ASCL redord:2311.002] — **DI**
VCAL-SPHERE: Hybrid pipeline for reduction of VLT/SPHERE data
 Contribution: Minor coding contribution (author 5/5)
- (1) Lillo-Box et al., 2022 [A&A, 667, A102] — **RV**
The KOBE experiment: K-dwarfs Orbiting By habitable Exoplanets. Project goals, target selection, and stellar characterization
 Contribution: Sample selection and Sect. 4.3 (author 8/36)

Book Chapters

- (1) Lillo-Box & **Balsalobre-Ruza** (to be published), Elsevier
Exoplanet Detection Methods. Chapter 8. Orbital Phase Modulations

3 Research experience

- **Predoctoral Researcher [PhD thesis]** CAB (Madrid, Spain)
 Search for co-orbital exoplanets in the *TRÖV* project Mar 2022 — Present [3 years]
 - **Predoctoral stay** at Liège University (Belgium) March/August 2025 [3 weeks]
 Collaboration with Dr. Valentin Christiaens on high-contrast imaging
 - **KOBE hackathons** CAB (Madrid), LAM (Marseille), GO (Genève)
 Three separate hackathons for RV analysis with the KOBE team May 2024, May & Nov 2025 [3 weeks]
 - **Predoctoral stay** at University California Santa Cruz (USA) Jun — Aug 2024 [2 months]
 Training in N-body simulations and collaboration with Dr. Nathaly Bathala's group
 - **Predoctoral stay** at ESO Santiago (Chile) Oct 2022 — Nov 2022 [1 month]
 Training in the analysis of ALMA observations with Dr. Itziar de Gregorio
- **Research Assistant** Universidad Autónoma de Madrid (UAM) (Spain)
 Project on exoplanet dynamics Dec 2021 — Mar 2022 [4 months]
- **Summer Internship** CEFA (Teruel, Spain)
 Characterization of atmospheric extinction at the Javalambre Observatory Jul 2021 — Sep 2021 [3 months]
- **Internship Trainee** CAB (Madrid, Spain)
 Support for Mars data analysis with the Rover Curiosity Feb 2021 — Jun 2021 [5 months]
- **Internship Trainee** CIEMAT (Madrid, Spain)
 Characterization of the LST-1 Cherenkov telescope camera Feb 2019 — Feb 2020 [1 year]

4 Coding

Python

- **6-year** experience as daily user
- **54-hour** online courses
- **Codes** available in [GitHub](#):
 - *moca* — RV simulator of planetary signals including activity with Gaussian processes. Focused on preparing observing proposals and injection recovery for sensitivity analysis [to be published]
 - VCAL-SPHERE — Collaborator in a reduction, pre-processing and post-processing SPHERE pipeline
 - KOBEsim — Bayesian algorithm to plan RV observations to boost the detection of exoplanets

Machine Learning & AI

- **240-hour** course by Samsung Innovation in the Polytechnic University of Madrid (UPM) in machine and deep learning
- **18-hour** course organized by IPARCOS group at UCM applied to physics and astrophysics
- Application of Random Forest algorithms in my **BSc thesis** with Python

Additional experience

- **Languages:** Matlab, R, IDL, Stan, SQL, HTML
- **SO:** macOS, Linux and Windows
- **Some astrophysics-relevant packages/programs:**
 - **Visualization:** carta, DS9
 - **RV:** radvel, kima, serval, raccoon, KOBESim, moca
 - **DI:** VCAL, VIP, star-hopping, CASA, irdap, LIFESim
 - **T:** lightkurve, wotan, batman
 - **General:** astropy, emcee, bayev, george

5 Observational experience

Visitor mode observations

- **HARPS-N [6 nights]** in the TNG (La Palma Observatory) [Jun 2024]
- **ESPRESSO [1 night]** in the VLT (ESO, Paranal Observatory) [Feb 2024]
- **HARPS [6 nights]** in the 3.6 m (ESO, La Silla Observatory) [Aug 2022]

Accepted observing programs

- **PI programs:**
 - (8) **SPHERE [2 h]** non-coronagraphic star-hopping IRDIS + IFS (ESO, Paranal Observatory) [P116]
 - (7) **GRAVITY [4 h] & SPHERE [2 h]** VLT-UT dual-field on-axis mode & non-coronagraphic star-hopping IRDIS + IFS (ESO, Paranal Observatory) **[DDT P115]**
 - (6) **CARMENES [39 h]** (CAHA Observatory) [25A]
 - (5) **CHEOPS [40 orbits]** (ESA) [AO-5]
 - (4) **CARMENES [44 h]** (CAHA Observatory) [24B]
 - (3) **SPHERE [4.5 h]** coronagraphic IRDIS DPI star-hopping (ESO, Paranal Observatory) [P113]
 - (2) **CARMENES [11 h]** (CAHA Observatory) [23B]
 - (1) **CARMENES [22 h]** (CAHA Observatory) [22B]
- **Co-PI programs:**
 - (8) **CARMENES** (CAHA Observatory) – PI: C. Cifuentes
 - (7) **NIRPS & HARPS** (ESO, La Silla Observatory) – PI: J. Lillo-Box
 - (6) **ESPRESSO** (ESO, Paranal Observatory) – PI: J. Lillo-Box
 - (5) **ESPRESSO** (ESO, Paranal Observatory) – PI: J.P. Faria
 - (4) **CHEOPS** (ESA) – PI: Jorge Lillo-Box
 - (3) **CARMENES** (CAHA Observatory) The KOBE experiment – PI: J. Lillo-Box
 - (2) **CAFOS** (CAHA Observatory) – PI: A. Castro-González
 - (1) **CAFÉ** (CAHA Observatory) – PI: J. Lillo-Box

6 Dissemination

Seminars

- (5) Geneva Observatory
The TROY project: Do co-orbital exoplanets really exist?

Genève (Switzerland)
Apr 2025

(4) Liège University	Liège (Belgium)
<i>The TRÖY project: Do co-orbital exoplanets really exist?</i>	Mar 2025
(3) ESA exoplanet team	Madrid (Spain)
<i>The TRÖY project: Do co-orbital exoplanets really exist?</i>	Oct 2024
(2) ESO headquarters	Santiago (Chile)
KOBESim. Bayesian observing strategy algorithm for planet detection in RV surveys	Nov 2022
(1) ESO headquarters	Santiago (Chile)
PiGs. Planets in Giant stars	Aug 2022

Conferences

• Talks

(10) Detection and Dynamics of Exoplanets	Coimbra (Portugal)
<i>Looking closer at PDS 70: Tracing Trojan dust?</i>	Jul 2025
(9) Madrid-Area Exoplanet Science Meeting at ESAC	Madrid (Spain)
<i>The KOBE experiment. First results</i>	Dec 2024
(8) Bay Area Exoplanets Meeting	Santa Cruz, CA (USA)
<i>The TRÖY project: co-orbital exoplanets and how to find them</i>	Jul 2024
(7) Other Worlds Laboratory Summer Program 2024	Santa Cruz, CA (USA)
<i>Developing Moca to compute mock RVs induced by exoplanets</i>	Jul 2024
(6) Keep uPhDated CAB	Madrid (Spain)
<i>Genesis of Trojans in PDS 70 b potentially hunted with ALMA</i>	Dec 2023
(5) Toward Other Earths III	Porto (Portugal)
<i>Genesis of Trojans in PDS 70 b potentially hunted with ALMA</i>	Jul 2023
(4) PhD meetings UCM	Madrid (Spain)
<i>Exotrojan planets. Missing pieces of planetary formation</i>	Mar 2023
(3) Keep uPhDated CAB	Madrid (Spain)
<i>Exotrojan planets. Missing pieces of planetary formation</i>	Dec 2022
(2) Madrid-Area Exoplanet Science Meeting at ESAC	Madrid (Spain)
<i>In the search for exotrojan planets</i>	Oct 2022
(1) Spanish Society of Astronomy XV at IAC	Tenerife (Spain)
<i>The KOBE experiment - KOBESim: improving RV detection through efficient scheduling</i>	Sep 2022

• Posters

(6) Extremely Precise Radial Velocities 6 (EPRV 6)	Porto (Portugal)
<i>Exotrojans around low-mass stars: A search with a RV clock</i>	Jun 2025
(5) Extremely Precise Radial Velocities 6 (EPRV 6)	Porto (Portugal)
<i>KOBE-1: A new multi-planetary system around a late K dwarf</i>	Jun 2025
(4) Complex Planetary Systems II	Namur (Belgium)
<i>Witnessing the genesis of exotrojans in PDS 70 [Awarded]</i>	Jul 2023
(3) ESLAB 2023 ESTEC [Remote participation]	Noordwijk (Netherlands)
<i>Towards completing extrasolar systems with the TRÖY project</i>	Mar 2023
(2) Scientific Meeting of the Spanish Society of Astronomy XV	Tenerife (Spain)
<i>The KOBE experiment - KOBESim: improving RV detection through efficient scheduling</i>	Sep 2022
(1) LATSIS 22 ETH [Remote participation]	Zurich (Switzerland)
<i>Co-orbital worlds as possible bullets of giant impacts</i>	Sep 2022

Technical talks

• PyCoffee – Python short seminars series	CAB, Madrid
<i>Regular expressions for a more efficient work</i>	Jan 2025
<i>Start doing your own Python modules</i>	Mar 2024
• Journal Club – Paper presentation series	CAB, Madrid
<i>Reviewing: DSHARP VII. The Planet-Disk Interactions Interpretation</i>	Nov 2024

Outreach and media

- Nacional Radio and international written interviews on my research
Onda Regional, Radio Extremadura, ABC, Science, Space among others
- Press-releases on my research work
ESO, CAB, CAHA
- Collaborator in the outreach YouTube channel of my research group
Remote Worlds Lab
- First-authored article for the Revista Española de Física
Acercándonos al desierto donde residen los mundos habitables (to be published, V40)
- Four outreach talks in schools
Hidden planets in the immense Universe (Madrid, Apr 2025)
- Co-writer in a scientific blog and instagram content
Le dijo un Quark al Cosmos (Jun 2020 - Jan 2022) [Not available anymore]

Courses on communication skills

- Course on TV, Radio and News Conferences for researches
Organized by the Spanish National Radio and television (RTVE) and CSIC (17 h, Feb 2025)
- Writing better proposals and Papers workshop
Organized by Dr. Artie Hatzes
- Speak in public and present work efficiently course
Organized by UCM

Organizing experience

- **LOC** of a workshop in ESAC
Exploring Tatooine and beyond: Circumbinary planets with ESA missions
Madrid (Spain)
Dec 2025
- **Co-organizer** of PyCoffees series at CAB
See [Github repository](#)
Madrid (Spain)
Nov 2023 – Present
- **SOC** of a mini-symposium within the XV meeting of the Spanish Society of Astronomy
Here and There: Synergies between Solar System and Exoplanetary Research
Granada (Spain)
Jul 2024